IES · GenAI for Engineers

Lab 3.1 · Prompt
FMEA Document and RPN Table

How this prompt is built

This prompt follows **R.I.S.E.** — **R**ole sets the perspective, **I**nput names exactly which evidence may be used, **S**teps make the work traceable, and **E**xpectations define what finished looks like. The independent check at the end is yours, not Copilot's.

	Block	What it does
R	**Role**	Sets the perspective Copilot answers from.
I	**Input**	Names exactly which evidence may be used, and forbids the rest.
S	**Steps**	Makes the work traceable so a reviewer can follow it.
E	**Expectations**	Defines what finished looks like, and what must never appear.
YOU	**Independent check**	Your hand calculation. This one is not Copilot's job.

Every block below is labelled with its framework letter. Keep the labels when you reuse this prompt — they are what make it repeatable.

Copy this prompt into Copilot in Excel

Select the named table first, then open Copilot and paste everything below.

R ROLE · Role

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Act as an FMEA facilitator supporting a competent engineer. You draft; you never approve.
```

I INPUT · Input

```
Use only the named Excel table FMEAInput on the FMEA_Input sheet, and the classroom scale on
the Rating_Guide sheet. Do not use outside knowledge and do not add rows.
```

S STEPS · Steps

```
1. Preserve all four supplied rows and every supplied S, O and D rating exactly as given.
2. Calculate RPN = Severity x Occurrence x Detection for each row and show the arithmetic,
for example 7 x 5 x 4 = 140.
3. Rank the four rows from highest to lowest RPN.
4. Separately flag every row with Severity 9 or higher under the heading
ESCALATE FOR SEPARATE REVIEW, whatever its RPN.
5. Insert a clustered column chart of RPN by failure mode ID.
6. For each row give one evidence request and one practical follow-up action.
```

E EXPECTATIONS · Expectations

```
Return a table with the supplied columns plus RPN and Severity flag, then the ranked list,
then the escalation list, then the actions. Show the RPN formula once in words. Do not invent
incidents, causes, owners or dates. Do not change a rating. Do not state that the skid is safe
or fit for service. Note that a high Detection number means POOR detectability.
End with the unresolved questions and the words DRAFT FOR REVIEW.
```

YOU INDEPENDENT CHECK · Independent check

```
Verify FM-01 by hand: 7 x 5 x 4 = 140. Confirm FM-04 is escalated on Severity 9 even though
its RPN is the lowest of the four.
```

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.